Traffic Engineering

METI

Lab 2 Report [EN]

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1 Introduction

The objective of this laboratory assignment was to implement and analyze IP routing mechanisms using Cisco routers within the GNS3 simulation environment.

The experiments performed throughout the laboratory focused on dynamic routing protocols, routing behavior, path selection mechanisms, and routing convergence.

The laboratory activities included:

- IPv4 subnetting using VLSM;
- router interface configuration;
- OSPF routing configuration;
- routing table analysis;
- OSPF metric manipulation;
- connectivity verification;
- link failure simulation and convergence analysis.

The network topology consisted of five routers (R1 to R5) and two end devices (PC1 and PC2), interconnected through multiple redundant paths.

Part A — OSPF Network Configuration

2 Network Topology

The network topology implemented in GNS3 is presented below.

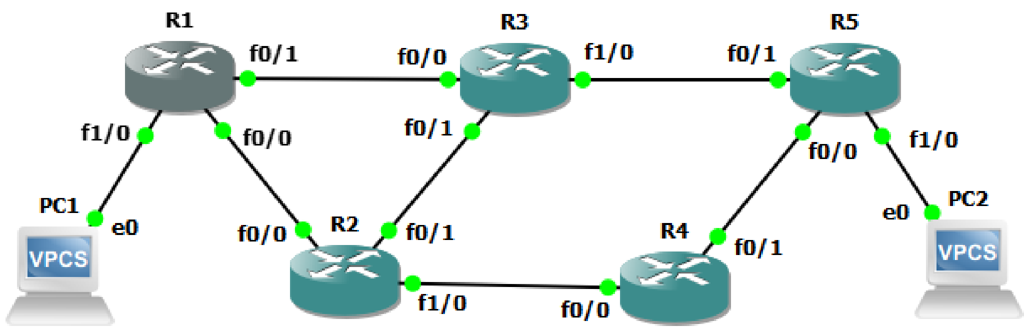

Figure 1: General network topology

The topology contains five routers interconnected through multiple redundant paths, allowing OSPF to dynamically calculate alternative routes in case of failures.

Additionally:

- PC1 is connected to R1;
- PC2 is connected to R5.

2.1 Broadcast Domains

Each physical link corresponds to an independent subnet/broadcast domain.

The network contains:

- 6 router-to-router subnets;
- 2 LAN subnets (PC1-R1 and PC2-R5).

Therefore, the topology contains a total of:

8 broadcast domains

3 IPv4 Addressing Plan

The network address block assigned was:

192.168.5.0/24

Using VLSM, the address space was segmented into multiple /30 subnets.

Link	Subnet	IPs Used
R1-R2	192.168.5.4/30	.5 / .6
R1-R3	192.168.5.8/30	.9 / .10
R2-R3	192.168.5.12/30	.13 / .14
R2-R4	192.168.5.16/30	.17 / .18
R4-R5	192.168.5.20/30	.21 / .22
R3-R5	192.168.5.24/30	.25 / .26
PC1-R1	192.168.5.0/30	.1 / .2
PC2-R5	192.168.5.28/30	.29 / .30

Table 1: IPv4 addressing plan

4 Router Configuration

Each router interface was configured manually.
Example configuration used on R1:

```
interface FastEthernet1/0
ip address 192.168.5.2 255.255.255.252
no shutdown

interface FastEthernet0/0
ip address 192.168.5.5 255.255.255.252
no shutdown

interface FastEthernet0/1
ip address 192.168.5.9 255.255.255.252
no shutdown
```

The command below was used to verify interface status:

```
show ip interface brief
```

All interfaces reached the `up/up` operational state.
The same configuration methodology was applied to all routers in the topology.

5 OSPF Configuration

OSPF was configured using process ID 1.

The following commands were applied:

```
router ospf 1
network 192.168.5.0 0.0.0.255 area 0
```

After configuration, OSPF neighbor relationships were successfully established.

The command used to verify adjacency status was:

```
show ip ospf neighbor
```

Routers reached the FULL adjacency state, confirming successful synchronization of OSPF databases.

All routers were configured within OSPF Area 0.

6 Routing Table Analysis

The routing table was inspected using:

```
show ip route
```

Routes learned dynamically through OSPF were identified with the letter:

$$O$$

Initially, multiple equal-cost paths existed for some destinations, demonstrating Equal Cost Multi-Path (ECMP) behavior.

Example:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                        via 192.168.5.6
```

This indicates that two paths with identical OSPF cost were available.

This demonstrated the capability of OSPF to support load balancing across equal-cost paths.

7 OSPF Cost Manipulation

To analyze OSPF path selection behavior, the cost of interface FastEthernet0/1 on R1 was increased.

Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After modifying the cost:

- OSPF recalculated the routing table;
- alternative lower-cost paths were selected;

- traffic was redirected automatically.

Traceroute confirmed the path modification.
Before the cost change:

$$R1 \rightarrow R3 \rightarrow R5$$

After increasing the interface cost:

$$R1 \rightarrow R2 \rightarrow R4 \rightarrow R5$$

This demonstrated OSPF metric-based path selection.
This experiment demonstrated how OSPF metrics directly influence path selection decisions.

8 Experimental Results

This section presents the experimental results obtained during the implementation and testing of the network.

8.1 Interface Verification

After configuring the interfaces, the following command was used to verify interface status:

```
show ip interface brief
```

Example output from R1:

```
Interface           IP-Address      Status      Protocol

FastEthernet0/0     192.168.5.5     up          up
FastEthernet0/1     192.168.5.9     up          up
FastEthernet1/0     192.168.5.2     up          up
```

The output confirmed that all interfaces were correctly configured and operational.

8.2 OSPF Neighbor Adjacency

The following command was used to verify OSPF neighbor relationships:

```
show ip ospf neighbor
```

Observed output:

```
Neighbor ID     Pri   State       Dead Time   Address         Interface

192.168.5.25      1   FULL/BDR    00:00:32    192.168.5.10    FastEthernet0/1
192.168.5.17      1   FULL/DR     00:00:33    192.168.5.6     FastEthernet0/0
```

The FULL state confirmed that OSPF adjacencies were successfully established and routing databases were synchronized.

8.3 Routing Table Analysis

The routing table was analyzed using:

```
show ip route
```

Example output:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                        via 192.168.5.6

O 192.168.5.24 [110/11] via 192.168.5.10

O 192.168.5.16 [110/11] via 192.168.5.6
```

Routes marked with the letter `O` correspond to routes learned through OSPF.

Initially, some routes had multiple equal-cost paths, demonstrating Equal Cost Multi-Path (ECMP) behavior.

8.4 OSPF Cost Manipulation

To influence path selection, the OSPF cost of interface FastEthernet0/1 on R1 was increased.

Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After changing the cost, the routing table changed accordingly.

Updated routing example:

```
O 192.168.5.12 [110/20] via 192.168.5.6

O 192.168.5.24 [110/21] via 192.168.5.6

O 192.168.5.20 [110/21] via 192.168.5.6
```

This confirmed that OSPF selected the path with the lowest accumulated metric.

8.5 Connectivity Verification

Connectivity between end devices was tested using ICMP echo requests.

Command executed:

```
ping 192.168.5.30
```

Observed output:

```
!!!!!
Success rate is 100 percent (5/5)
```

This confirmed successful communication between PC1 and PC2.

Traceroute was also performed:

```
traceroute 192.168.5.30
```

Observed route before OSPF cost manipulation:

```
1 192.168.5.6
2 192.168.5.14
3 192.168.5.26
4 192.168.5.30
```

After increasing the OSPF cost, the selected route changed:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

This demonstrated successful dynamic route recalculation.

8.6 Link Failure and OSPF Convergence

To evaluate convergence behavior, one of the active interfaces was disabled.
Command executed:

```
interface FastEthernet0/0
shutdown
```

OSPF immediately detected the failure:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17 on FastEthernet0/0
from FULL to DOWN
```

The routing table was automatically recalculated and traffic was redirected through alternative paths.
Traceroute after failure:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

The interface was restored using:

```
no shutdown
```

OSPF adjacency was re-established successfully:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17
from LOADING to FULL
```

This demonstrated:

- automatic fault detection;
- dynamic rerouting;
- successful OSPF convergence.

Part B — TBD

This section will contain the experiments and analysis developed in Part B of the laboratory assignment.

Part C — TBD

This section will contain the experiments and analysis developed in Part C of the laboratory assignment.

9 Conclusion

This laboratory assignment successfully demonstrated the deployment and analysis of an OSPF-based IP network using Cisco routers in GNS3.

The implementation of VLSM enabled efficient IPv4 address allocation across all network segments. OSPF dynamically exchanged routing information and automatically selected optimal paths according to interface metrics.

Several experiments were performed to evaluate OSPF behavior under different network conditions, including:

- routing table analysis;
- OSPF cost manipulation;
- connectivity verification;
- link failure simulation;
- routing convergence analysis.

The obtained results demonstrated:

- successful end-to-end connectivity;
- dynamic path recalculation;
- automatic failover capabilities;
- fast OSPF convergence.

Overall, this assignment provided practical experience with dynamic routing protocols and fundamental traffic engineering concepts.